Rajesh Sudam

Ireland | +353 899848071 | rajeshsudam21@gmail.com | LinkedIn | GitHub | Medium | Stamp 1G

Professional Summary

Master's Graduate in Computer Science with a strong interest in Artificial Intelligence, Machine Learning, and Data Science. A curious and analytical problem-solver with experience across research and industry environments, motivated by turning complex data into meaningful insights and solutions. Eager to begin a career in AI and Data Science, contributing to impactful projects while continuing to develop as a well-rounded data and AI professional.

Education

University of Galway, Galway, Ireland
MSc Computer Science (AI & Cybersecurity) — First Class Honours
CMR Technical Campus, Hyderabad, India
Bachelor of Technology in Information Technology — First Class Honours

Research & Relevant Projects

Lossy Text Compression using Genetic Algorithms (AICS 2025, peer-reviewed) GitHub

- Designed and published a novel lossy text compression framework combining Genetic Algorithms with GPT-2 for compression guidance, presented at AICS 2025.
- Evaluated compression quality using Sentence-BERT semantic similarity and Bits-Per-Character (BPC), balancing storage efficiency against content preservation.
- Independently researched, prototyped, and published the work while completing a full MSc course load and a teaching assistantship.

AI-Driven Claims Companion GitHub

- Applied Retrieval-Augmented Generation (RAG) techniques to ground LLM responses in domain documents and improve decision support.

Employee Performance Tracking System GitHub

- Built a full-stack, scalable web application (Angular, Spring Boot, PostgreSQL) with REST APIs, authentication, and role-based user dashboards.

Experience

Teaching/Research Assistant, Immersive Software Engineering (ISE) Sep 2026 – Present
University of Limerick, Ireland

- Deliver lectures and support teaching for third-year undergraduate students in the ISE department.
- Evaluate exam papers and provide structured, consistent feedback against marking criteria.
- Assisting research projects under Professors and Collaboration with Lero
- Assist students in writing and debugging code, building their problem-solving and analytical skills.

Graduate Teaching Assistant Jan 2025 – Apr 2025
University of Galway, Ireland

- Conducted weekly lab sessions, assisting students with programming, debugging, and structured problem-solving.
- Collaborated with professors to deliver course content and foster teamwork in a fast-paced academic environment.
- Evaluated assignments and delivered code reviews, applying close attention to detail and consistent quality standards.

Associate Analyst Sep 2023 – Dec 2023
GlobalLogic, India

- Performed systematic inspection and labelling of image and video datasets, ensuring data quality and compliance with strict annotation standards before handover to AI/ML teams.
- Investigated and resolved data non-conformances by producing detailed defect reports and liaising with cross-functional teams, preventing recurrence through clear documentation and root-cause analysis.

- Managed and resolved tickets related to data quality issues, applying structured triage to close corrective actions within agreed SLAs.

Intern Software Developer Jan 2023 – Aug 2023
Hitachi Vantara, India

- Built and maintained independent front-end and back-end modules integrated into enterprise applications.
- Designed REST APIs and microservice components to improve application scalability and maintainability.
- Collaborated with UX designers and product stakeholders to translate user requirements into working interfaces.
- Debugged, tested, and code-reviewed production systems within an Agile delivery team.

Technical Skills

- **Programming:** Python, SQL, Java, JavaScript/TypeScript
- **Machine Learning & Deep Learning:** Supervised/unsupervised learning, neural networks, model training and evaluation, hyperparameter tuning, PyTorch, TensorFlow/Keras, scikit-learn
- **NLP & Generative AI:** Transformers, Hugging Face, Retrieval-Augmented Generation (RAG)
- **Data Analysis & Visualisation:** Pandas, NumPy, Matplotlib, exploratory data analysis, data cleaning, dataset annotation and quality assurance
- **MLOps & Tools:** Git/GitHub, Docker, CI/CD, Linux, Azure, AWS, GCP
- **Software Engineering:** REST APIs, Spring Boot, Angular, Node.js, PostgreSQL
- **AI Security & Responsible AI:** Cybersecurity foundations, malware and intrusion detection analytics, societal impact of AI, OWASP Top 10.

Certifications

- SQL Intermediate (HackerRank)
- IBM zSystems Cybersecurity Insights (IBM)
- Microsoft SC-900 – Security, Compliance, and Identity Fundamentals
- CC – Certified in Cybersecurity (ISC2)
- SOC Level 1, Security Engineer, Cyber Security 101 (TryHackMe)

Achievements & Extracurricular Activities

- Winner – National AI Challenge 2026, TechIreland: collaborated in a multidisciplinary team to develop an MVP and pitch an AI solution at the national finals.
- Published researcher: Lossy Text Compression using Genetic Algorithms – presented at AICS 2025.
- Participant, IdeasLab – received President's Award for Innovation and Entrepreneurship.
- Event Coordinator, Cybersecurity Society – University of Galway (Oct 2024 – May 2025).

REFEREES

References Available Upon Request